8.1 Computer Science Fundamentals

8.1.1 Algorithm

A set of instructions that are defined to perform a specific computing task. This can be something as simple as performing addition or something more complex like modifying an image in an image editor. Programmers spend a majority of their time creating algorithms that are then debugged to ensure that they work properly and efficiently. The primary goal is to create efficient algorithms that use computing resources like RAM and CPU time with the maximum efficiency. Obviously, when an algorithm is badly written, it will be a drag on performance.

8.1.2 Analogue

Anything that is read or accessed as a continuous stream is experienced as analogue. Humans perceive everything in analogue—as a continuous stream of data and information to our senses. Similarly, the grooves in a record player or the magnetic data stored on an audio or video tape is typically a representation of analogue data. In comparison, digital is an approximation of the audio or video signal. This means that analogue data is more accurate than digital data, but due to some inherent advantages with the digital format, analogue is being slowly eliminated.

8.1.3 Bespoke

In technology circles, a bespoke solution or system is something that is custom-developed to meet the requirements of a specific need. This can refer to both hardware and software. Typically an organisation would go for bespoke software if there is no off-the-shelf software package that can meet their needs. Software developers will then build a customised system to meet the requirements. Typically, bespoke solutions are expensive, and with the increase in commoditisation in the software market, more and more solutions are coming out-of-the-box with customisation features rather than being 100 per cent bespoke. The cases where only a bespoke solution will fit are rapidly shrinking from the general marketplace and are getting focused into niche segments.